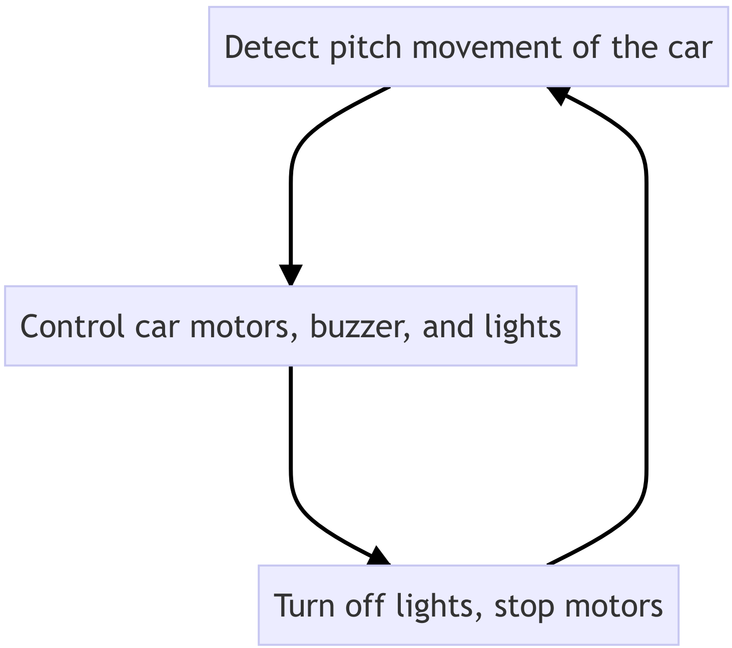
11. Chasing Frogs

Function Description

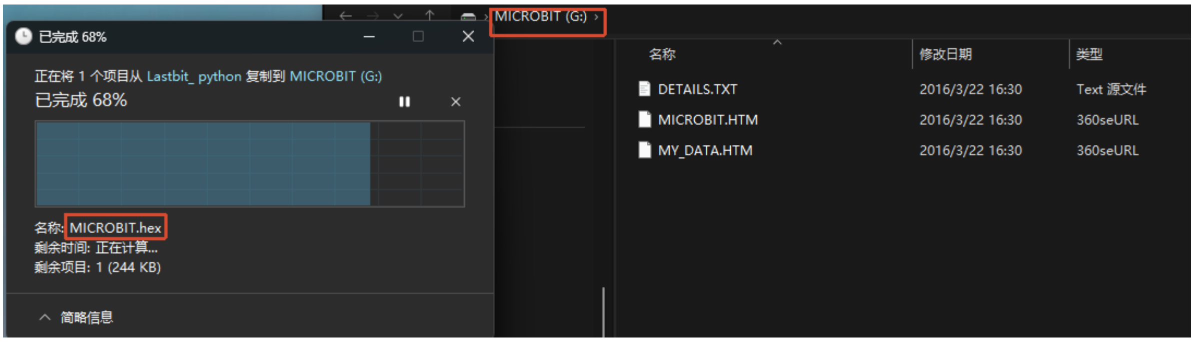
In this section, we will learn to use the accelerometer sensor to detect tilt and achieve the "Chasing Frogs" effect. When the car is detected to tilt forward, it will make a sound and jump forward, while lighting up lights of random colors.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



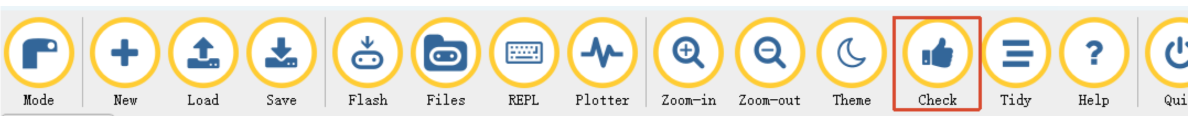
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

   Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

4. Now you can click `Check` to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.

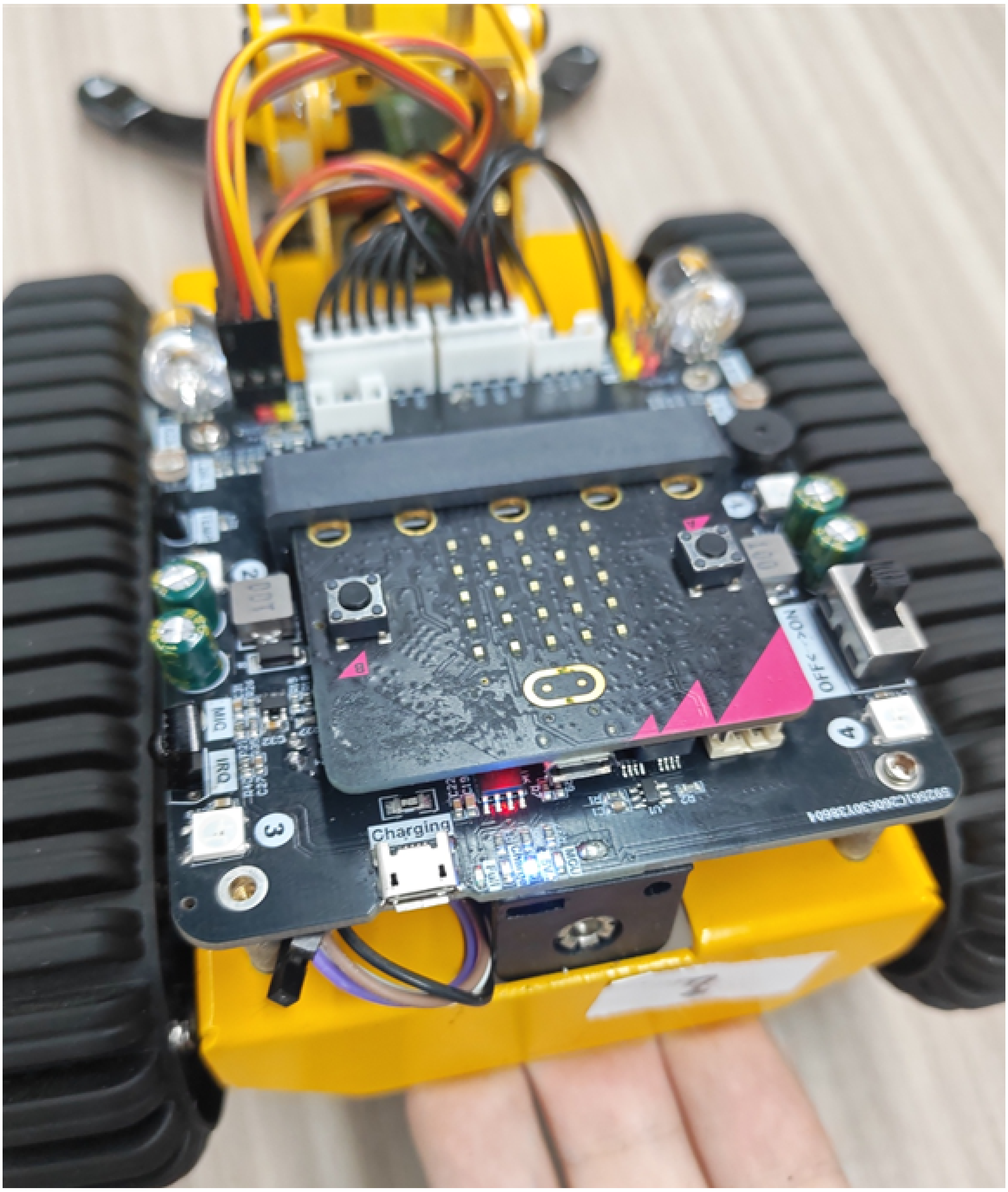
6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 11. Chasing Frogs
from microbit import *
from lastbit import LASTBIT
import music
import random

Lastbit = LASTBIT()

while True:
    # Read pitch angle change, two methods provided, use the more stable one, you can also adjust the threshold yourself
    angle_change = accelerometer.get_y()
    # angle_change = Lastbit.get_pitch()

    # Tilt forward
    # angle_change: modify here to adjust the detection angle of the car
    if angle_change < -10:
        # Generate random color
        r = random.randint(0, 255)
        g = random.randint(0, 255)
        b = random.randint(0, 255)

        # Set lights
        Lastbit.all_lights_set_color(r, g, b)

        # Play sound
        music.pitch(294, 100)
        sleep(20)
        music.pitch(220, 140)
        sleep(60)

        # Move forward
        Lastbit.car_control('FORWARD', 50, 1000)
        Lastbit.stop_motor()

        # Turn off lights
        Lastbit.all_lights_off()

    sleep(50)
```

Experimental Phenomenon

The car will detect its current angle information. Lift the rear of the car and then release it. The car will light up its headlights and move forward, and the buzzer will also make sounds simulating the croak of a frog. The angle threshold (angle_change), pitch, and tempo parameters can be adjusted according to actual needs.



Notes

It is recommended to conduct the experiment on a flat surface.